

Multi-omics Informed Drug Response Prediction for Precision Medicine

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ABSTRACT

Motivation: Treating cancer in patients is hard within precision oncology due to the heterogeneity in cancer cells. Developing predictive methods that estimate drug-responses in cancer cell lines can improve our understanding of treatment effectiveness. The goal of this study is to explore the fundamental and data-driven limitations of the existing advanced learning methods for drug-response prediction for precision oncology.

Approach & Findings: We examine current state-of-the-art methods that utilize multi-omics patient data and/or signaling pathways to predict drug response across various cell lines. Our analysis identifies that models that learn molecular and topological manifolds demonstrate richer representation of drug molecules, and better efficacy of drug-response prediction. We also identify the core design principles that are necessary for developing an effective computational framework. Future work will explore various options in the architecture design principles and attempt to improve drug-response prediction, especially when multi-omics data is not present for a patients.

INTRODUCTION

Precision oncology faces significant challenges due to intra-tumor heterogeneity and lack of reliable cancer biomarkers for accurate diagnosis and treatment [9]. This genetic and environmental diversity leads to variable drug efficacy among patient cohorts. Additionally, wet-lab drug screening remains costly and time-consuming. Therefore, computational models that can predict therapeutic responses from patient-specific genetic profiles hold immense potential in personalized cancer treatments.

Machine learning and deep learning algorithms have been widely applied in creating drug-response prediction models. While these models perform better when multi-omics is present, often perform suboptimal for low-omics overlapping scenario across patients. A core bottleneck is to design methods for encoding high-dimensional and sparse single/multi-omics data. Additionally, cross-omics optimal feature identification remains a key challenge in the drug-response prediction.

To address these gap, we studied fifteen recent state-of-the-art methods, focusing on the multi-omics genetic profiles of patients and their drug responsiveness in treatments, to explore the fundamental and data-driven limitations of the existing advanced learning methods. Our comprehensive analysis reports two-fold findings:

- (i) Methods that learn molecular and topological manifolds demonstrate richer representation of drug molecules, and better efficacy of drug-response prediction.
- (ii) We identify the core design principles for developing an effective computational framework.

Following these principles, we reproduce a framework that **combines latent representation of high-dimensional multi-omics data, locality features, and spatial relations of drug-tissue interactions**. Our implementation achieves comparable performance with exiting platform across multiple metrics. The network preserves the most salient features across multi-omics data and drug molecular representations, especially when complete data points are not available.

METHODS AND MATERIALS

We perform a comprehensive literature review of recent deep learning models encompassing multi-omics usage and drug-responsiveness. While these approaches share similar predictive objectives, they evaluate performance across diverse open-source cohorts (summary in **Table 1**).

Citation	GDSC	CCLL	TCGA	PDX	LINCS
[1]	Yes	Yes	No	No	No
[8]	Yes	Yes	No	No	No
[6]	Yes	Yes	No	No	No
[11]	Yes	Yes	No	No	No
[4]	Yes	Yes	No	No	No
[10]	Yes	Yes	No	No	No
[5]	Yes	Yes	No	No	No
[12]	Yes	No	No	No	No
[13]	Yes	No	Yes	No	No
[2]	Yes	No	Yes	No	No
[9]	Yes	No	Yes	Yes	No
[3]	No	No	No	Yes	Yes
[7]	No	No	Yes	No	No

Table 2. Datasets used by each cited recent state-of-the-art drug-response model.

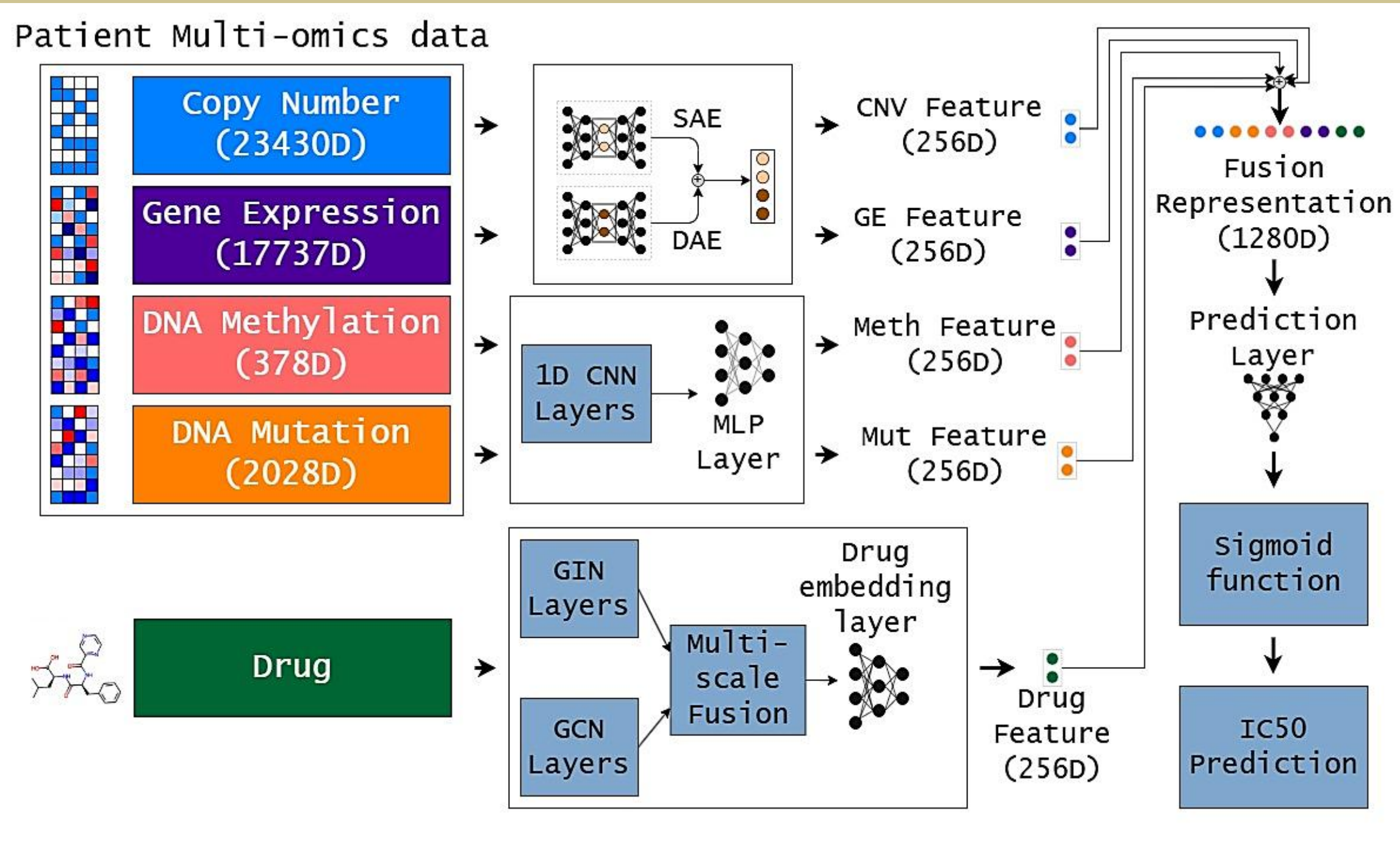


Figure 1. A schematic overview of a patient's omics profile-based drug-response prediction model.

METHODS AND MATERIALS (cont.)

Our principled architecture have four modules:

- (i) P1: A representation module for dense omics profiles, i.e gene expression (mRNA) and copy number variations (CNV).
- (ii) P2: A representation module for sparse omics profiles, i.e DNA methylation and mutation.
- (iii) P3: A topological encoder to capture drug-molecular interaction features.
- (iv) P4: A final prediction layer.

Following this, we replicate the DeepMoDRP [5] which combines a multiscale Graph Convolutional Network (GCN) with a fusion of Sparse Autoencoder (SAE) to predict drug response in brain cancer.

Our implementation incorporates one drug feature and four cell line features as inputs, with the IC50 value serving as the predicted output. We implemented four modules:

- i. An autoencoder (AE)-based **P1** module
- ii. A convolutional neural network (CNN)-based **P2** module
- iii. A graph neural network (GNN)-based **P3** module
- iv. A fully connected layer **P4** module that combines all features to predict IC50 values.

In the provided DeepMoDRP implementation, a standard autoencoder was used instead of the sparse autoencoder described in the original framework. Therefore, our replication follows the implementation's use of a conventional autoencoder.

Dataset sources: We adopt the same datasets used by DeepMODRP [5], particularly Genomics of Drug Sensitivity in Cancer (GDSC) [14] and Cancer Cell Line Encyclopedia (CCLE) [15] databases and screening data from The Cancer Genome Atlas Program (TCGA) for brain cancer.

Evaluation: We evaluate our implementation in terms of standard performance metrics: root mean square error (RMSE), Pearson's correlation coefficient (PCC), Coefficient of Determination (R²), and mean absolute percentage error (MAPE).

RESULTS AND DISCUSSION

Our implementation performs on par with the original DeepMoDRP framework (**Table 2**).

We also compare our implementation with six state-of-the-art models. Our principled architecture-based implementation achieves superior performance across all four metrics (**Table 2**). This also demonstrates the reproducibility and efficacy of the DeepMoDRP [5] framework.

Despite the fact that methods use similar benchmark datasets (GDSC and CCLE), differences in experimental settings, cancer scope, evaluation protocols, and data missing statistics should be considered when interpreting results.

Method	RMSE	PCC	R ²	MAPE, %
Ours	0.0472	0.9449	0.8923	8.52
DeepMoDRP	0.0480	0.9454	0.8933	8.16
MCMVDRP	0.0485	0.9434	0.8898	8.46
PASO	-	0.9425	0.8838	-
HyDRP-LG	0.0584	0.8881	0.7886	-
GraTransDRP	0.0239	0.9341	-	-
TopDr	-	0.9423	-	-
MCA	0.0457	0.9284	0.8619	-

Table 1. Performance comparison of drug-response prediction models reported across previous studies which use benchmark datasets GDSC and CCLE.

CONCLUSION AND FUTURE WORKS

Current deep learning frameworks are becoming increasingly effective in drug-response prediction because of richer feature representation. Within our principled design, we demonstrate that tailoring encoder topologies to specific multi-omics modalities, extending beyond sparse versus dense data categorizations, is essential for capturing distinct biological signals. Therefore, future works will explore various architecture within the principled design. Additionally, highlighting the field's dependence on cell-line-based multi-omics data which are conducted *in vitro* experiments, which might limit clinical transferability when in treating actual tumors. Hence, our future endeavor will focus on multi-source patient data to enhance generalizability and clinical applicability.

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